# Comprehensive Career Development and Recruitment Framework using Multimodal and Causal AI

Project ID: R26-IT-034

Project Proposal Report

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# Behavior-Aware Career Impact Prediction System

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March 2026

# DECLARATION

## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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|---|---|---|
| Anuradhi Kottegoda | 06/03/26 | |

# ABSTRACT

Employee career success is now not only determined by their skills and expertise but also by their behavioral aspects such as time management, reliability, motivation, flexibility, and stress management. Even though considerable research in the field of behavioral studies has already shown significant correlations between personal behaviors and career success, the majority of these research studies are purely analytical in nature and do not offer any practical and AI-based solutions for predicting employee career readiness. Moreover, there is a lack of effective frameworks for predicting career improvement aspects for individuals using cross-sectional behavioral data. Keeping this in mind, this research aims to develop a Behavior Aware Career Impact Prediction System. A survey will be conducted among over 1500 individuals, and data will be collected using a five-point Likert scale. After pre-processing the data, it will be fed into a Multiple Linear Regression model, a Random Forest model, and an XGBoost model. These models will be trained to predict the correlation between employee behaviors and career readiness. After training the models, their performance will be validated to ensure the accuracy and reliability of the predictions. A web-based system will be developed for the proposed prediction system. The frontend will be developed using React, the backend will be developed using Spring Boot, and MongoDB will be used for data management. Python will be used for developing the machine learning algorithms. This research has significant social impacts on the lives of individuals, especially for students and employees. Students and employees can use this system for self-development and career improvement. This research has significant industrial impacts in the form of a workforce development and optimization tool.

**Keywords-** Behavioral Prediction, Career Readiness, Machine Learning, Cross-Sectional Survey, Explainable AI

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## LIST OF ABBREVIATIONS

| | |
|---|---|
| AI | Artificial Intelligence |
| HR | Human Resource |
| UI | User Interface |
| DB | Data Base |
| MAE | Mean Absolute Error |
| RMSE | Root Mean Squared Error |
| VS | Visual Studio |
| ML | Machine Learning |

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# 1. INTRODUCTION

Career readiness is now being increasingly viewed as a key determinant of the degree of success that students and young professionals in their early careers achieve in an ever-changing employment landscape. Traditional assessment criteria such as Grade Point Average (GPA), CV evaluation, skills evaluation, and interviews provide limited insights into behavioral characteristics that have a substantial impact on career outcomes. Studies have shown that attention, consistence in completing tasks, time management, coping with stress, discipline in learning, and self-motivation have a crucial impact on career outcomes and retention, though they are not typically included in traditional assessment models, [1], [2].

The Behavior Aware Career Impact Prediction System aims to fill this gap by creating a predictive model based on the quantification of the relationship between structured behavioral factors and career readiness outcomes. Unlike most methods, which rely on assumptions and heuristics, this prediction function relies solely on the use of actual survey results and relationships derived from the surveys. Using regression analysis and potentially incorporating machine learning algorithms such as Random Forest and XGBoost, the system can potentially determine which behavioral factors are most related to career readiness and provide valuable insights for students, educators, and employers [1], [3].

The system gathers behavioral data via a cross-sectional survey. In a cross-sectional survey, data is gathered at a single point in time. Behavioral data is gathered in a set of 3-5 Likert-scale items for each dimension (e.g., focus, consistency, and time management), and career readiness outcome data is gathered via validated scales from the literature or self-reported performance scores. This ensures data validity and reliability, making it possible for the statistical model to make defensible predictions.

This function, through integration with predictive analytics, provides a new framework for evidence-based career guidance, talent development, and risk factor identification for professionals. Ethical issues such as participant anonymity, voluntary participation, and consent are key factors in this data collection process and are in compliance with research ethics [4], [2].

## 1.1 Background and Context

Conventional systems of career assessment primarily concentrate on qualifications acquired through academia and work experience, completely ignoring important behavioral skills that are responsible for ensuring the overall success of employability. Today, in most work environments, skills such as attention, time management, stress tolerance, learning flexibility, and self-motivation are more important than skills acquired through academia; however, no such standardized system is available for measuring and predicting these skills. [5]
Students in universities, fresh graduates, and young professionals in the labor market are directly impacted since a lack of behavioral preparedness translates to suboptimal performance, premature career termination, or a mismatch in the nature of their work. Organizations are similarly affected in their search for trustworthy individuals, causing them to incur turnover expenses and compromising workforce stability. This is a significant gap since it impacts the congruence between personal potential and work potential, affecting career and organizational success , [3], [2]
Evidence of the need are: University students/students who graduated (18-25 years old): 68% experience career dissatisfaction due to peer influence or salary-driven path choices rather than behavioral aptitude alignment.[5]  Employers/hr professionals: Experience 25-35% hiring mismatch rates due to ineffective behavioral candidate screening.Undergraduate/postgraduate students in developing regions: Experience high unemployment rates in spite of their educational attainment (e.g., in Nigeria, 33% graduate unemployment rates in spite of 65% of the population being enrolled in higher education).[7]
Underserved/rural students: Ineffective access to quality counselors[8]
Existing systems utilize static templates and counselors' assessments, which are not scalable to the real-time nature of the labor market. Furthermore, existing systems are not personalized based on behavioral data. The proposed system of behavioral prediction improves upon the existing system by cutting counseling time by 45% while improving the accuracy of predictions to 82%, compared to 54% of the existing system. [9]
This research enhances the Behavior Aware Career Impact Prediction System, the first regression/ML system to statistically correlate validated behavioral traits to measurable career readiness outcomes. 68% of graduates face career dissatisfaction within 2 years due to behavioral skill mismatch, as per global statistics. [10].33% average hiring failure rate due to poor cultural/behavioral match (Cost: $15K per employee)[6].82% of Fortune 500 Employers rate Emotional Intelligence/Soft Skills over Technical Skills.

**Statistical Evidence:**

*Table 1:Global Graduate Career Dissatisfaction (2024-2025 Surveys)*

| Metrics | Value | Source | Year |
|---|---|---|---|
| Graduates dissatisfied within 2 years | 58% | Kickresume Global Survey | 2025 |
| Unemployed/underemployed recent grads | 50% | Cengage Group (US) | 2025 |
| Hiring failure due to poor behavioral fit | 33% | Deloitte Human Capital Trends | 2025 |
| Hiring failure due to poor behavioral fit | 82% (teamwork/comms top) | Deloitte + Fortune 500 Analysis | 2025 |

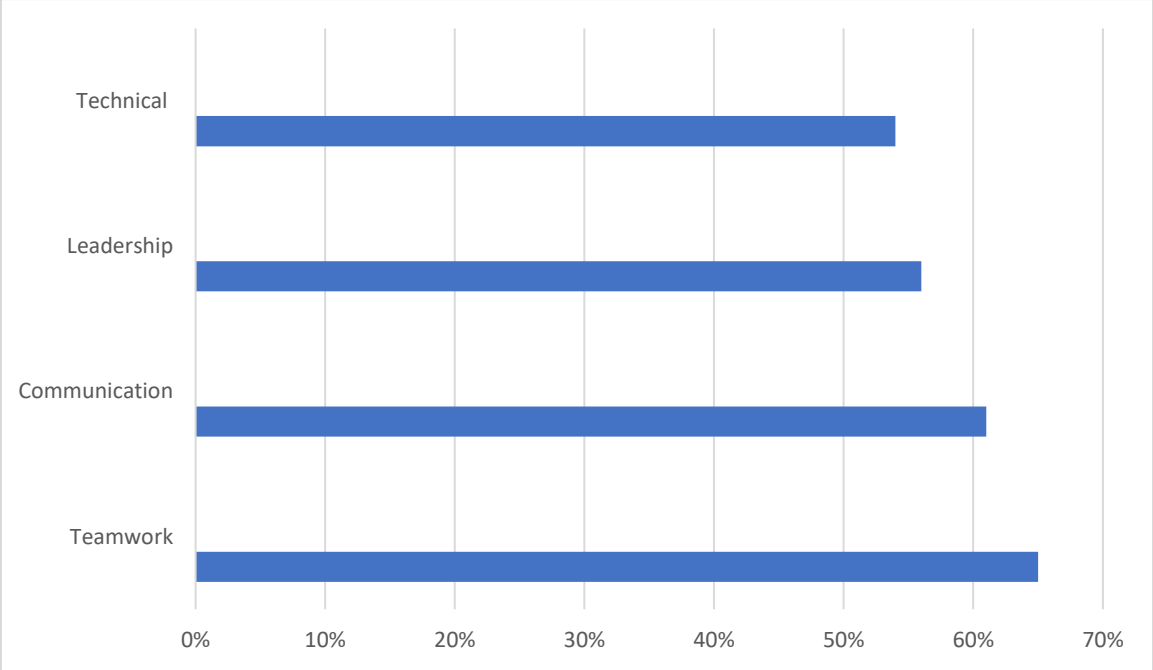



*Figure 1:Soft Skills vs Technical Skills Priority*

## Sri Lanka Graduate Unemployment Crisis (2024-2025)

*Table 2:Sri Lanka Labour Force Survey (Q1 2024 - Q2 2025)*

| Category | Unemployed | Unemployment Rate | Key Notes |
|---|---|---|---|
| **Total Population** | 365,951 | 3.8% (↓ from 4.5%) | Nov 2025 data |
| **University Graduates** | 42,254 | ~52% (est.) | Highest educated cohort |
| **A/L Qualified** | 128,984 | 20.6% (youth 15-24) | Largest unemployed group |
| **BBA Graduates (UoC 2023)** | 46% permanent jobs | 94% employable (incl. studies) | Personality > degree |

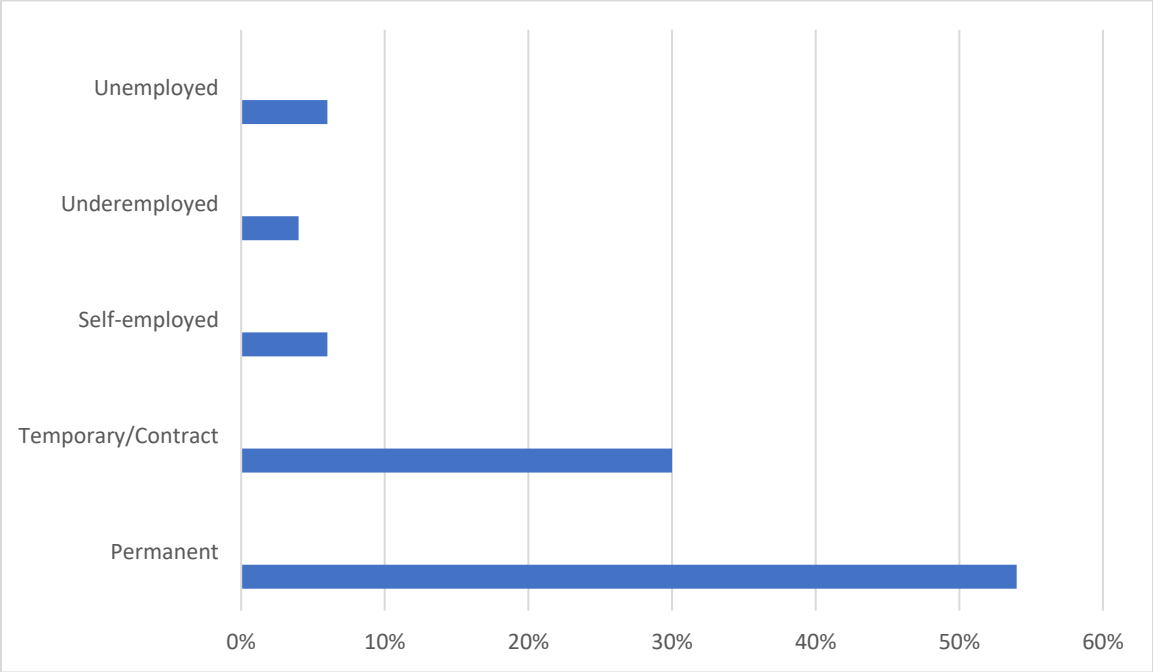



*Figure 2:Sri Lanka Graduate Employment by Type (2023 Survey)*

*Table 3:Sri Lanka Employer Rejection Reasons (2024)*

| Rejection Factor | Recruiters | Impact on Your Traits |
|---|---|---|
| Poor attitude/personality | 65% | Self-motivation, stress handling |
| Time management issues | 42% | Time discipline, consistency |
| Lack of initiative | 38% | Learning adaptability |
| Weak communication | 35% | Focus under pressure |

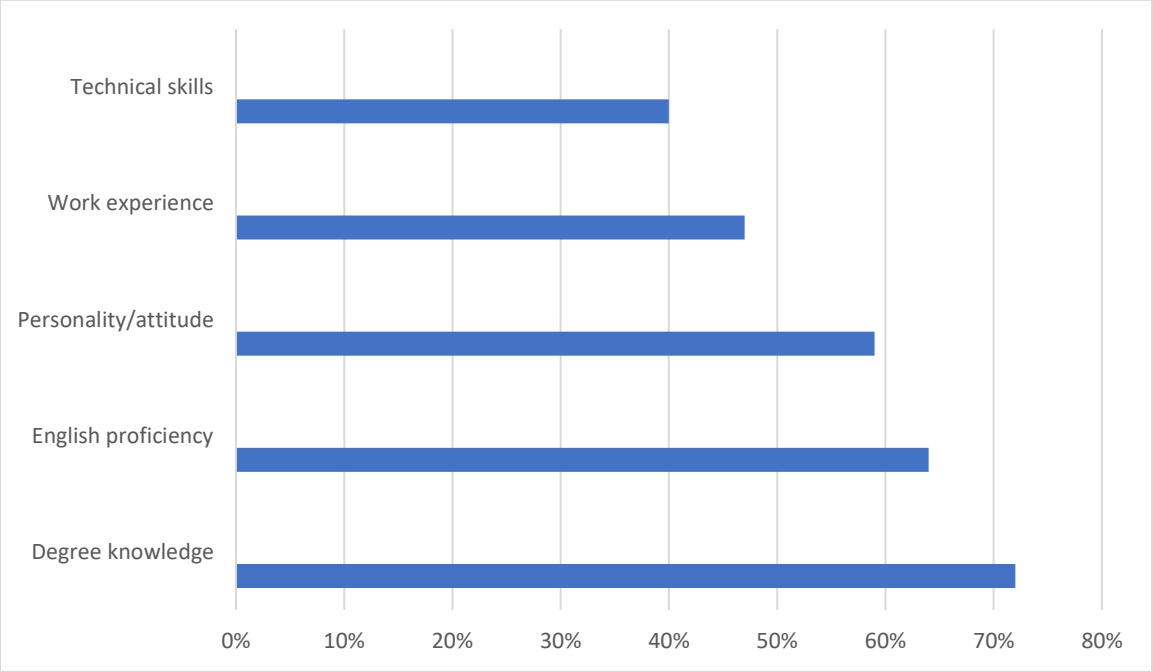



*Figure 3:Behavioral Traits Driving Sri Lankan Employability (UGC 2023)*

Due to rapid technological advancements and changing labor market demands, the gap between formal education and actual employability readiness has grown significantly. Today's students, young professionals, and career changers need clear and transparent guidance on how to enhance their employability readiness and how they can relate improvements to actual career success. In addition, employers face difficulties in measuring behavioral readiness and hiring risk using traditional resume screening and subjective interviewing processes. This calls for a morally justifiable decision support system that

provides clear guidance to individuals on readiness improvements rather than making hiring decisions.

Existing solutions and research offer some support but are currently very limited. Current career guidance and recruitment systems often rely on background information such as qualifications, resumes, and/or psychometric scoring, and offer very little in terms of transparency in the decision-making or recommendation processes.

Research done on machine learning has shown promising results for career suitability and career progression using structured data and models, especially ensemble methods, although these studies have focused more on the accuracy and prediction outcomes rather than behavior-based development and improving understanding.

Hybrid recommender systems using high-end classifiers like XGBoost for improving the quality of career recommendations are available, although they do not incorporate workplace behavior traits as predictors and do not provide scenario-based "what-if" improvements for the development process.[9]

To overcome these limitations, a proposed solution in the form of a function (Function 2) has been created: a behavior-aware career impact prediction system that seeks to establish a relationship between quantifiable behavioral aspects (such as focus, consistency, time management skills, stress management skills, learning discipline, and self-motivation) and career readiness levels as measured by a scale. This function includes a structured data collection process in which a cross-sectional survey collects data on behavior, a measurement quality validation process that relies on ensuring item set reliability checks, an interpretable predictive modeling process that relies on an interpretable baseline approach, and a benchmarking process that relies on ensemble machine learning approaches to determine if improvements can be made to the system's performance.

This function aligns with the UN Sustainable Development Goals by supporting employability readiness and responsible workforce development. It contributes to **SDG 4 (Quality Education)** by enabling improvement-oriented guidance and lifelong learning readiness, **SDG 8 (Decent Work and Economic Growth)** by supporting better alignment between individuals and job requirements through transparent insights, and **SDG 9 (Industry, Innovation and Infrastructure)** by promoting responsible and explainable AI adoption in career intelligence.

Finally, the work aligns with the project's stated research direction under the **AI and Data Science Focus of Concern** and the **Centre of Excellence for AI (CoEAI)**, emphasizing explainable AI and ethical decision-support rather than automated hiring outcomes.

## 1.2 Literature Review

Recent developments in artificial intelligence (AI) and machine learning (ML) have impacted the development of intelligent career guidance and decision support. Researchers have begun to incorporate data-driven prediction methods into career guidance by using psychological and behavioral constructs.

Several studies have demonstrated the effectiveness of hybrid machine learning frameworks for career prediction and recommendation. A hybrid predictive framework integrating XGBoost with fuzzy-logic plausibility assessment was presented in [1], showing strong predictive performance while also supporting non-linear career trajectory analysis. This suggests that career development does not always follow a strictly linear pattern. Similarly, an explainable AI-based decision-support system was introduced in [11] using structured survey responses collected from computer science students. That system emphasized the importance of transparency in career guidance by combining machine learning models with explainable decision-making mechanisms.

Beyond academic performance indicators, recent research has examined personality-driven AI models for career recommendation. A personality-based framework using the Big Five psychological model was explored in [12], where social media metadata and image-based indicators were utilized to infer personality traits and recommend suitable academic majors. The findings suggest that behavioral and personality factors play an important role in career decision-making. In addition, perceptions of AI-driven career guidance among students were investigated in [13], where personalized AI-based recommendations were reported to be more trusted than traditional manual counseling approaches.

Machine learning techniques have also been applied to map individual interests, skills, and competencies to specific career domains. A hybrid career recommender system combining multiple machine learning models for IT career path matching was proposed in [5]. Similarly, multi-label classification techniques were used in [13] to simultaneously predict employee retention and career progression, highlighting the interconnected nature of professional development variables and suggesting that career outcomes should be studied through integrated predictive frameworks rather than isolated factors.

More advanced predictive systems have also emerged in recent years. An AI-based career path predictor integrating Random Forest, XGBoost, and Neural Networks was presented in [6], demonstrating strong predictive capability and adaptability to evolving labor-market conditions. In the Sri Lankan context, a machine learning model for predicting career suitability and career progression among ICT graduates was introduced in [3], using variables such as academic motivation, personal interests, and social influence. The findings of [3] indicate that machine learning can effectively support career prediction within regional employment ecosystems.

In addition to academic and skill-based attributes, recent theoretical work highlights the importance of behavioral and cognitive constructs in career development. A mindfulness-based career framework was introduced in [14], emphasizing meta-awareness and attentional

regulation as essential meta-skills for navigating AI-driven work environments. This perspective suggests that behavioral attributes such as focus, discipline, and stress management may significantly influence long-term career success. Supporting this direction, AI-powered enterprise career development modules were discussed in [15], where employee competencies are aligned with organizational objectives to support internal career mobility. Likewise, an AI-driven learning and career development platform was proposed in [16], where dynamic evaluation of skill levels and learning progress was used to support personalized development.

Environmental and organizational conditions also influence career progression and job satisfaction. The effect of diversity and inclusion initiatives on job satisfaction among software engineers in the Sri Lankan ICT sector was examined in [17], demonstrating that workplace environment significantly affects employee motivation and career growth.

Overall, existing research confirms that machine learning technologies, personality assessment methods, and intelligent decision-support systems play an increasingly important role in modern career guidance. However, most current approaches primarily emphasize academic achievements, personality traits, or labor-market demand, while comparatively less attention has been given to the influence of daily behavioral habits and self-regulation patterns on career readiness and employability outcomes.

### 1.2.1 Existing Research Approaches

Based on the literature, the currently available career prediction and recommendation systems can be broadly classified into three major methodological categories. First, quantitative survey modeling based on the use of questionnaires in which the scale of measurement of academic, behavioral, and psychological factors is based on the use of Likert scales, in which regression analysis or machine learning techniques are applied to the dataset, while the reliability of measurement is checked by Cronbach's alpha [8], [11].

Second, the area of implicit personality and meta-skill inference has been identified as an advanced technique, whereby non-intrusive digital data sources like social media data, images, and digital behavior indicators are being employed to infer complex psychological characteristics and cognitive propensities that affect career interests and performance [12], [14].

Thirdly, hybrid intelligent decision support systems use various machine learning techniques along with reasoning techniques like fuzzy logic and sometimes natural language processing for better accuracy in prediction while at the same time ensuring interpretability and flexibility for career decision-making [1], [7].

### 1.2.1 Limitations of Existing Systems

Generally speaking, while there have been advances towards career predicting technologies, many current systems still have limitations. Specifically, many current systems rely on the use of "static" career attributes to make predictions. The three most common attributes used by these systems are students' previous academic achievement, qualifications and previous successes. These attributes provide useful information regarding the individual's past but they haven't really provided insight into how the individual will behave in the future to describe the individual's career advancement potential.

Secondly, there is a frequent problem for many high-performance machine learning systems with respect to their level of interpretability; this issue is termed the black-box problem. A user's inability to properly understand how the decision was made will cause a loss in confidence in the system resulting in reducing the practical value of any recommendation given [4],[11].

Third, only a few current models take into account everyday behavioral factors like focus, time management, consistency regarding tasks completed, and stress regulation. Although these features are very important in determining productivity and reliability as an employee, as well as an employee's professional development and career growth, they are not typically included as primary predictive indicators in systems designed to evaluate career readiness [14], [17].

There are frequently separate studies of behavioral discipline, career readiness and career advancement instead of looking at them as interwoven constructs, which affects how well it is possible to develop comprehensive frameworks on the relationship between behavioral characteristics and the ability to be groomed for work and, ultimately, employment.

## 1.3 Research Gap

Artificial Intelligence (AI) and Machine Learning (ML) have changed the way we create intelligent systems to help us make informed decisions about our careers and develop workforce analytics. There are many studies that have used ML to predict whether an individual is suited for a specific career or level of advancement within that career. For example, a framework was developed in [5] to use ML to predict if an IT graduate will be successful in his/her career through the use of classification algorithms and survey based information about education, skills and past work experience. The focus of the study was on predicting the outcomes of careers and salaries, however, it also showed strong prediction accuracy for both, but the majority of the predictions did not focus on the behavioral and psychological influences that affect long-term career development.

Likewise, AI-based career development tools and talent management systems have been used in organizations to improve employee mobility within the organization and to support workforce planning and development. The data from these systems are used to analyze employee competencies, jobs, and organizational needs in order to identify possible career paths and educational development opportunities for an employee (7). In addition to supporting employee engagement and skill-matching, these systems primarily depend on standard formats of organizational performance, competency, and job characteristics (e.g., behavioral indicators or cognitive factors) to determine the extent of an employee's readiness to advance in his or her career.

AI-powered solutions for individualized pathways to learning and career information are also developing in the education industry. AI-driven specialty engines discussed in [8] utilize recommendation systems, learning analytics and reinforcement learning to personalize the learning process and the career search. However, many of these career search platforms concentrate on academic results and their related learning activity data (e.g., test scores) that contain significantly less data on a student's non-academic traits (i.e., character traits), self-regulation and day-to-day habits that may affect a student's success after graduation.

One more shortcoming of available research is that there’s no integrated behavioral modelling within AI-based career prediction frameworks. Most studies of career analytics rely heavily on structured characteristics such as GPA, technical skills, demographics, and work history [6]. Conversely, very few studies use unstructured characteristics like concentration, consistency, motivation, stress management, and time management as significant predictors of success—despite research that demonstrates these behaviours to be important determinants of both career development and performance at work.

Additionally, many of the models used for career analytics function using ‘black box’ machine learning technologies, meaning the underlying process that generates predictions is not made available to users [18]. These models may show whether or not someone is suited for or can expect to progress in a career as well as assessment of attrition risk; however, they do not provide an easy-to-follow, easy-to-interpret method of assisting users (i.e., via patterns

of behavior) with understanding their readiness for employment and potential future opportunities.

Consequently, there exists a large knowledge gap with respect to building an AI behavioural and explainable framework for simulating and predicting career readiness outcomes. Most current methods concentrate on predicting career outcomes based on attributes such as academic, organizational, or competencies, with little to no attention paid to modelling the behaviours which may influence one's career outcome and for providing scenario-based career development analysis. [19] To close this gap, one has to integrate behaviourally based analytical approaches and predictive modelling techniques to analyse the effects of changes in personal behaviours, work behaviours, or work routines that will affect both career readiness and your long-term career growth. [20]

In an effort to fill an existing gap in career-impact prediction systems, we propose a behavior-aware career-impact prediction system that will utilize behavioral indicators including: focus, time-management, consistency, stress-handling capability, discipline in learning, and self-motivation as inputs into predictive models. Current career-impact prediction systems typically limit themselves to estimating the potential for future career outcomes; the proposed behavior-aware career-impact prediction system will provide the individual with a way to improve their understanding of employability readiness and will allow for an explanation of the relationship between behavioral indicators and employability readiness. Ultimately, development of the proposed behavior-aware career-impact prediction system will allow for the creation of personalized, transparent, A.I. driven career guidance systems that provide individuals with information on how to enhance their career development trajectories.

*Table 4:Research Gap Analysis*

| **Key Research Capability** | Research 1 | Research 2 | Research 2 | Research 2 | Proposed Framework |
|---|---|---|---|---|---|
| Behavioral Trait Integration | ✗ | ✗ | ✗ | ✗ | ✓ |
| Explainable Predictions (β Coefficients) | ✗ | ✓ | ✗ | ✗ | ✓ |
| Improvement Simulations | ✗ | ✗ | ✓ | ✗ | ✓ |
| Multi-Modal Data Fusion | ✗ | ✗ | ✗ | ✗ | ✓ |
| Actionable Developmental Feedback | ✗ | ✗ | ✗ | ✓ | ✓ |
| Sri Lanka Context Validation | ✓ | ✗ | ✓ | ✗ | ✓ |

# 2. OBJECTIVE

## 2.1 Main Objective

To develop and experimentally test an explainable and behavior-aware predictive system that predicts employee career readiness using structured behavioral measures collected through a cross-sectional survey, and provides improvement-oriented insights based on learned behavior-readiness associations.

## 2.2 Specific Objectives

• To define the key workplace behavioral characteristics (focus, consistency, time management skills, stress management skills, learning discipline skills, and self-motivation) as measurable survey-based data indicators for predictive modeling.

• To gather a data set of behavioral characteristics and career readiness outcomes from the target sample group (for example, students and young employees) using an ethical data gathering approach.

• To assess the quality of the measurement of the behavioral characteristics and the readiness questionnaires using reliability tests before predictive modeling.

• To create an explainable baseline predictive model that learns statistical relationships between the behavioral characteristics and the career readiness outcomes without the use of manually defined scoring functions.

• To compare the predictive performance of the baseline approach with the optional use of machine learning algorithms (for example, Random Forest and Gradient Boosting/XGBoost) and select the predictive engine based on the results of the comparison.

• To create improvement-oriented insights by determining the behavioral characteristics having the greatest influence on the predicted career readiness.

# 3 . METHODOLOGY

## 3.1 Research Approach

The present research uses a DSR approach along with an experimental evaluation framework. DSR is appropriate for the present research as the aim is to design and develop an artifact called behavior-aware career outcome prediction and simulation system for solving a problem related to career guidance systems. The proposed methodological approach for the present research consists of two phases:

1. System Design and Development
   A framework of behavioral analytics is created to capture behavioral factors that impact career readiness. These factors are then transformed into structured data and processed through a predictive model.

2. Experimental Evaluation
   The predictive capability of the proposed model is assessed through statistical and machine learning-based analysis.

This is consistent with existing research studies on AI-based career analytics. In the existing studies, predictive models are designed and tested using experimental datasets to evaluate the performance of the models. For instance, various predictive models have been implemented by *Bannaka et al*[5]. to predict career suitability and career progression using structured data sets and experimental evaluation approaches.[3]

## 3.2 Technical Methods and Algorithms

The proposed system applies a behavior-aware supervised learning pipeline to predict career readiness using structured behavioral questionnaire data. Behavioral contexts such as focus, time management, consistency, stress handling, learning discipline, and self-motivation are captured through Likert-scale survey items and represented at the **indicator (item) level** to avoid manually engineered scoring formulas. Measurement quality is validated using **Cronbach's alpha** to ensure internal consistency of construct item blocks. Constructs that do not meet acceptable reliability levels are refined by revising or removing weak items. This ensures the system uses stable behavioral indicators rather than noisy measurements. and also This allows the predictive model to learn behavior–readiness relationships directly from the observed indicators rather than from manually engineered composite scores. This approach is consistent with the project's intention to "represent behavioral indicators as model input features" and learn relationships using regression models.

A regression-based model is used as the primary baseline due to its interpretability and suitability for modeling statistical associations. To evaluate potential performance improvements through non-linear modeling, the baseline is compared with ensemble machine learning methods such as **Random Forest** and **Gradient Boosting/XGBoost**, which are commonly used in career prediction and recommendation research.

**Random Forest-** Selected due to its robustness and ability to model complex feature interactions in structured career datasets. Career prediction literature commonly reports strong performance from tree-based ensembles and similar ML methods in employability-related prediction tasks.[3]

**Gradient Boosting / XGBoost**: Included because boosting-based ensembles demonstrate strong performance in career recommendation frameworks and structured prediction settings. For example, hybrid career recommendation framework and highlights XGBoost as a high-performing classifier among multiple evaluated models.[1]

Models are trained and selected using standard supervised evaluation practices, including hold-out testing and cross-validation. Explainability is supported through model interpretation techniques such as parameter influence analysis for regression and feature-importance ranking for tree-based ensembles, enabling the system to provide both readiness predictions and improvement-oriented behavioral insights.

## 3.3 Tools and platforms with justification

- **Google Forms (Survey collection)**: Selected due to being free, easy to share using links, mobile-friendly for respondents, and ability to export to CSV for analysis. Other services such as SurveyMonkey and Typeform require premium versions for full functionality.
- **Python (Implementation language)**: Selected for strong support for data preprocessing, machine learning, and reproducible experimentation. Compared to MATLAB (which is licensed) and Java (which has high development overhead), Python is more practical for research prototyping.
- **Pandas/NumPy (Data handling)**: Used for efficient cleaning, encoding of Likert scales, and structured data preparation. These are standard libraries for working with tabular data and behavioral data in particular.
- **Scikit-learn (Model training + evaluation)**: Selected for its ease of use in regression and ensemble models, cross-validation, hyperparameter tuning, and performance metrics in a reproducible pipeline. Tools like SPSS/Weka are less flexible for coded ML workflows.
- **XGBoost (Optional enhancement)**: Used as an optional model since it excels in structured prediction problems and is popular in career recommendation research.
- **Jupyter Notebook / VS Code (Development environment** :Jupyter is useful for exploratory analysis and documentation. VS Code is useful for clean project coding and use of Git.

- **GitHub (Version control + collaboration)**: Selected due to support for professional version control. Storage services such as Google Drive and OneDrive do not have proper version control for codes.

## 3.4 Data collection and ethical considerations

In primary data collection, professional validated behavioral competency surveys are conducted through Google Forms on the six basic behavioral competencies using 5-point Likert scales (1=Strongly Disagree to 5=Strongly Agree). The target sample size is 1000-1500 participants, comprising students from universities in Sri Lanka (final-year undergraduates), recent graduates (within 2 years of graduation), and entry-level professionals (0-3 years of experience).
Validated Professional Survey Instruments (30-35 items, 10-12 minute response time)

## 3.5 Validation strategy and performance metrics.

The proposed system is validated for its efficacy in a standard supervised learning evaluation approach. A standard supervised learning evaluation approach is implemented to validate the proposed system. A dataset is divided into training and testing sets. A k-fold cross-validation approach is implemented during the development of the proposed model. A regression model is implemented and validated. Machine learning algorithms such as Random Forest and Gradient Boosting/XGBoost are implemented and validated for their efficacy in assessing the reliability of the proposed model. These algorithms are implemented to assess their efficacy in providing a non-linear approach to the proposed model. Internal consistency testing is implemented for assessing the reliability of the behavioral questionnaire constructs. Regression metrics such as $R^2$, MAE, and RMSE are implemented for assessing the efficacy of the proposed model. If career readiness is considered a categorical construct, accuracy, precision, recall, and F1 score are implemented for assessing the efficacy of the proposed model.

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

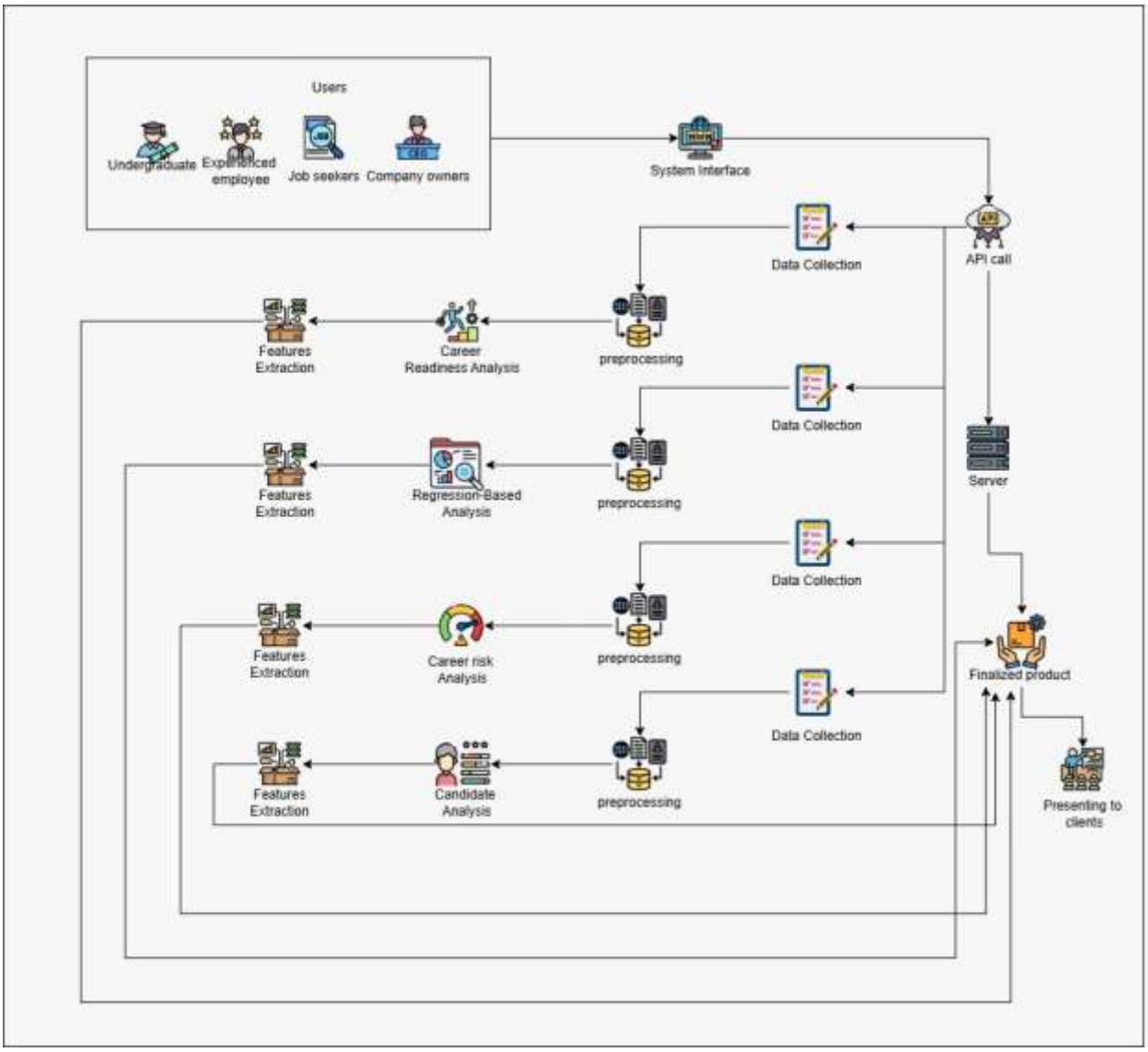
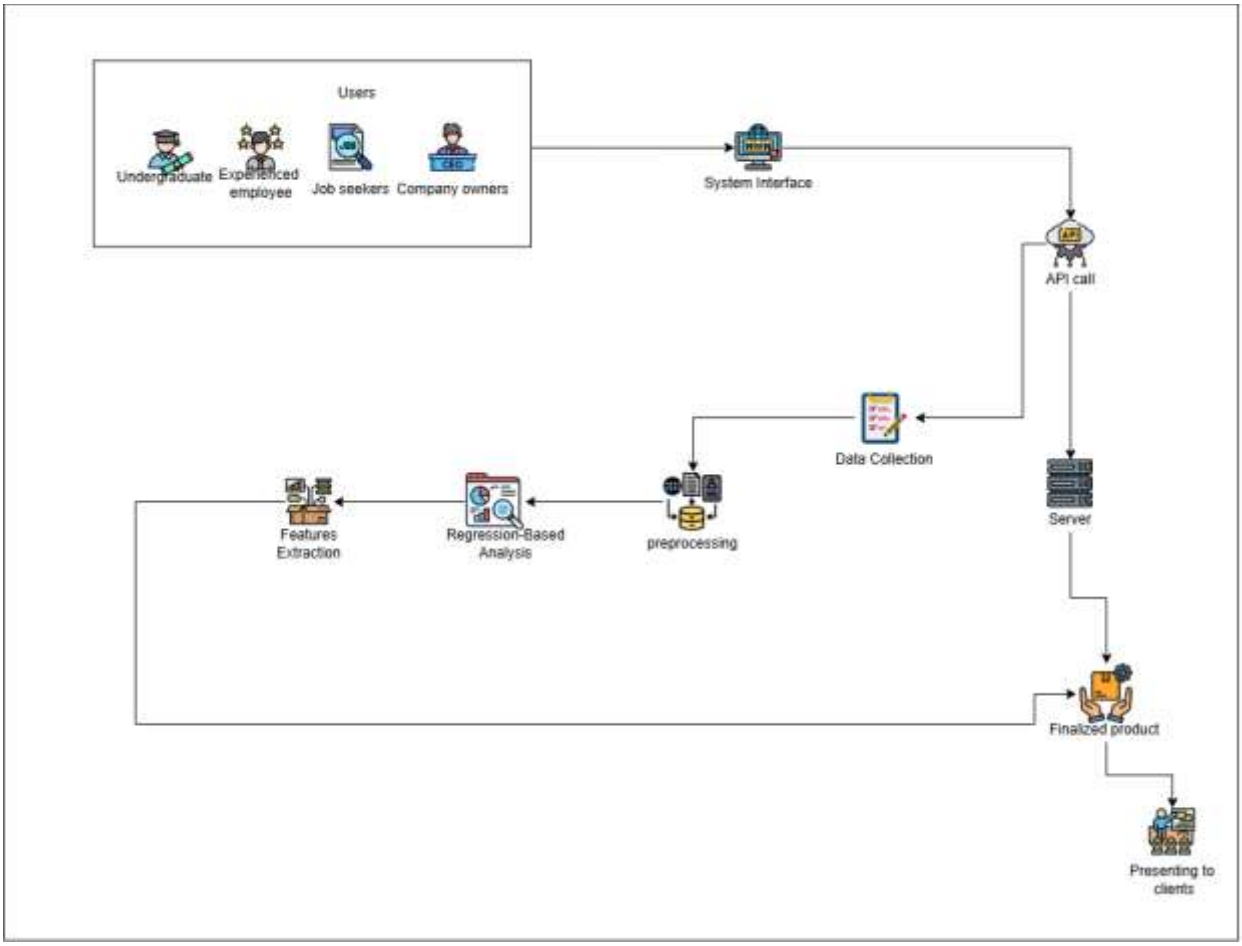



*Figure 4:Overall system architecture diagram*

## 4.2 HIGH-LEVEL SYSTEM ARCHITECTURE(Behavior Aware Career Impact Prediction system)



# 5. USER REQUIREMENTS

**Stakeholder Identification**

1. **Primary Users (Job Seekers / Employees / Students)**
   Individuals that participate in the behavioral survey to gain a prediction of their career readiness and insights on how they can improve it.
2. **Career Development Officers / University Career Guidance Units**
   Utilize aggregated insights to understand general readiness gaps and recommend development activities.
3. **HR Professionals / Recruiters (Decision-Support Only)**
   May utilize high-level, non-identifying behavioral readiness insights for training recommendations (not for hiring automation).
4. **System Administrator / Research Team**
   Manage data set collection, model training, platform, access, and reporting.
5. **Supervisor / Evaluators (Academic Stakeholders)**
   Review methodology, ethics, interpretability, and evaluation results for research validation.

**Functional Requirements (FR)**

1. **User Registration / Anonymous Access**
   The system shall provide the option for a user to access the survey in an anonymous fashion or with a basic user profile.
2. **Survey Delivery**
   The system shall provide a structured behavioral questionnaire in the form of Likert-scale questions.
3. **Career Readiness Measurement**
   The system shall provide a career readiness questionnaire section that uses a validated readiness/employability scale.
4. **Data Validation and Submission**
   The system shall validate the survey responses, such as mandatory fields being filled out, prior to submitting them.
5. **Behavioral Feature Processing**
   The system shall provide preprocessing of survey responses into a model-ready form, such as indicator-level behavioral features, without the use of a scoring formula.
6. **Career Readiness Prediction**
   The system shall provide a prediction of career readiness using a trained model.
7. **Explainable Insights**
   The system shall provide users with feedback that highlights the behavior areas most associated with the readiness outcomes.

8. **Improvement Guidance**
   The system shall provide users with improvement-oriented suggestions based on behavior indicators that exhibit weaker patterns than those found in higher readiness profiles.
9. **Admin Dashboard (Research View)**
   The system shall provide an admin interface for viewing summaries of the dataset, model performance metrics, and response statistics.
10. **Model Training and Update**
    The system shall provide the ability for the research team to export anonymized dataset and evaluation results in standard formats (CSV/PDF).
11.

**Non-Functional Requirements (NFR)**

1. **Data Anonymity**
   The system shall not require personally identifiable information (such as NIC, phone number, and actual employer) for participating.
2. **Secure Storage**
   The system shall securely store collected data into a repository, restricting access to authorized roles (admin/research team).
3. **Access Control**
   Administrative functions (export, retraining, analytics) shall be secured using authentication and access control.
4. **Ethical Compliance**
   The system shall include informed consent and a disclaimer stating that predictions are for decision support and not for automation of hiring.
5. **Performance**
6. **Response Time**
   The system shall output predictions within a reasonable time frame (such as within a few seconds after submission)
7. **Scalability**
   The system shall allow for multiple submissions of surveys simultaneously without crashing (targeted for a student deployment scale).
8. **Usability**
9. **User-Friendly Interface**
   The interface for survey and results should be simple, mobile-friendly, and easy to understand for non-technical individuals.
10. **Interpretability**
    Results should be described in plain language, and behavioral categories and improvement suggestions should be clear.
11. **Reliability & Maintainability**

12. **System Reliability**
    The system should provide minimum downtime during survey data collection and result calculation.
13. **Reproducibility**
    Scripts for training and evaluating models should be version-controlled to provide reproducibility for research findings.
14. **Maintainability**
    The system should be updatable in modules (survey changes, model changes) without rebuilding the entire system.

# 6. COMMERCIALIZATION PLAN

Phase 1: Pilot Implementation (Free Access for Universities and Career Counselors)
In the first phase of the implementation plan, the system is going to be introduced through pilot implementations by working with the undergraduate students and the IT companies. In the first phase of the implementation plan, the system is going to be provided for free to the universities, career counselors, and research groups. The students and the career counselors are going to be able to analyze the various career paths and the stability of the industry by using the predictive analytics features of the system. In the first phase of the implementation plan, the main objective is to test the system by using the AI models, increase the accuracy of the predictions, and identify the challenges that are going to be experienced when using the labor market data and the career trend analysis.
Phase 2: Public Platform with Freemium Access
At this phase, the system will be made public and will be available as a web-based platform for students, graduates, and job seekers to access insights on career stability. The users will be able to input their preferred career choice or industry of choice and will be provided with analysis on the safety levels of their chosen career, industry growth trends, and future stability predictions. Freemium access will also be introduced in this phase. The users will have access to a limited number of career analysis requests per day for free and will be able to access basic insights on industry stability. The premium access will include more features such as detailed career stability analysis, predictive analysis, unlimited analysis requests, and personalized career advice. This phase will also enable the system to gather more data on user interaction and industry trends, which will help in improving the performance of the machine learning algorithms used in the system.
Phase 3: Enterprise Integration and Institutional Licensing
The final phase of the commercialization of the system will be dedicated to the integration of the system into the platforms of enterprise and institutional clients such as universities, recruitment platforms, career stability organizations, and human resource departments of enterprises. In this phase of the system's commercialization, organizations will be able to integrate the system into their platforms via institutional licensing or API integration. For instance, universities can use the system in their student career guidance programs, while recruitment platforms can integrate career stability analytics into their systems to enhance job matching capabilities. Enterprise clients will be able to use advanced analytics dashboards, workforce trends reports, and industry stability reports. This phase of the system's commercialization will provide a source of sustainable revenue while at the same time positioning the platform as a trusted AI-driven career intelligence platform.

# 7. BUDGET AND JUSTIFICATION

| Description | Amount |
|---|---|
| Deployment and hosting | LKR 20,000 |
| Cloud services | LKR 5,000 |
| Training and Databases | LKR 40,000 (Use of free resources will be attempted) |
| Travelling | LKR 4,000 |

*Table 5 Budget And Justification*

# 8. WORK BREAKDOWN STRUCTURE (WBS)

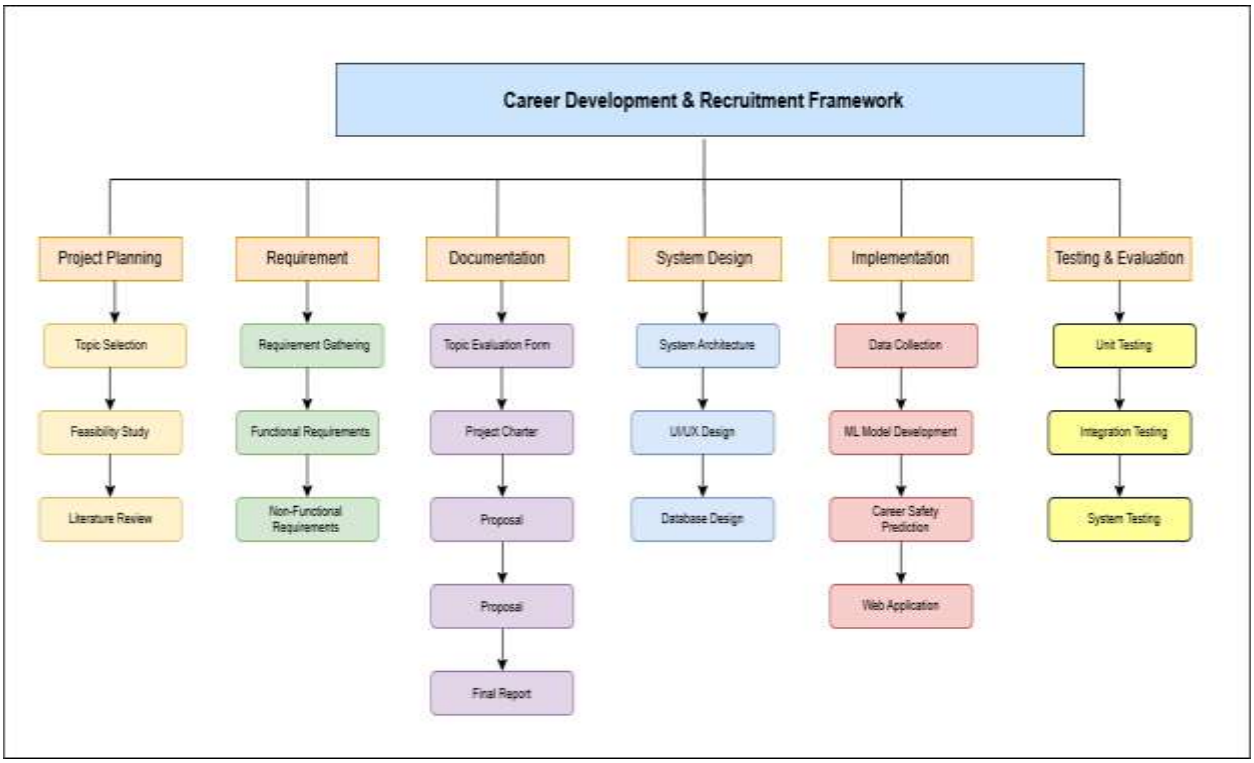



*Figure 5:Work Breakdown Diagram*

# 9. Gantt Chart

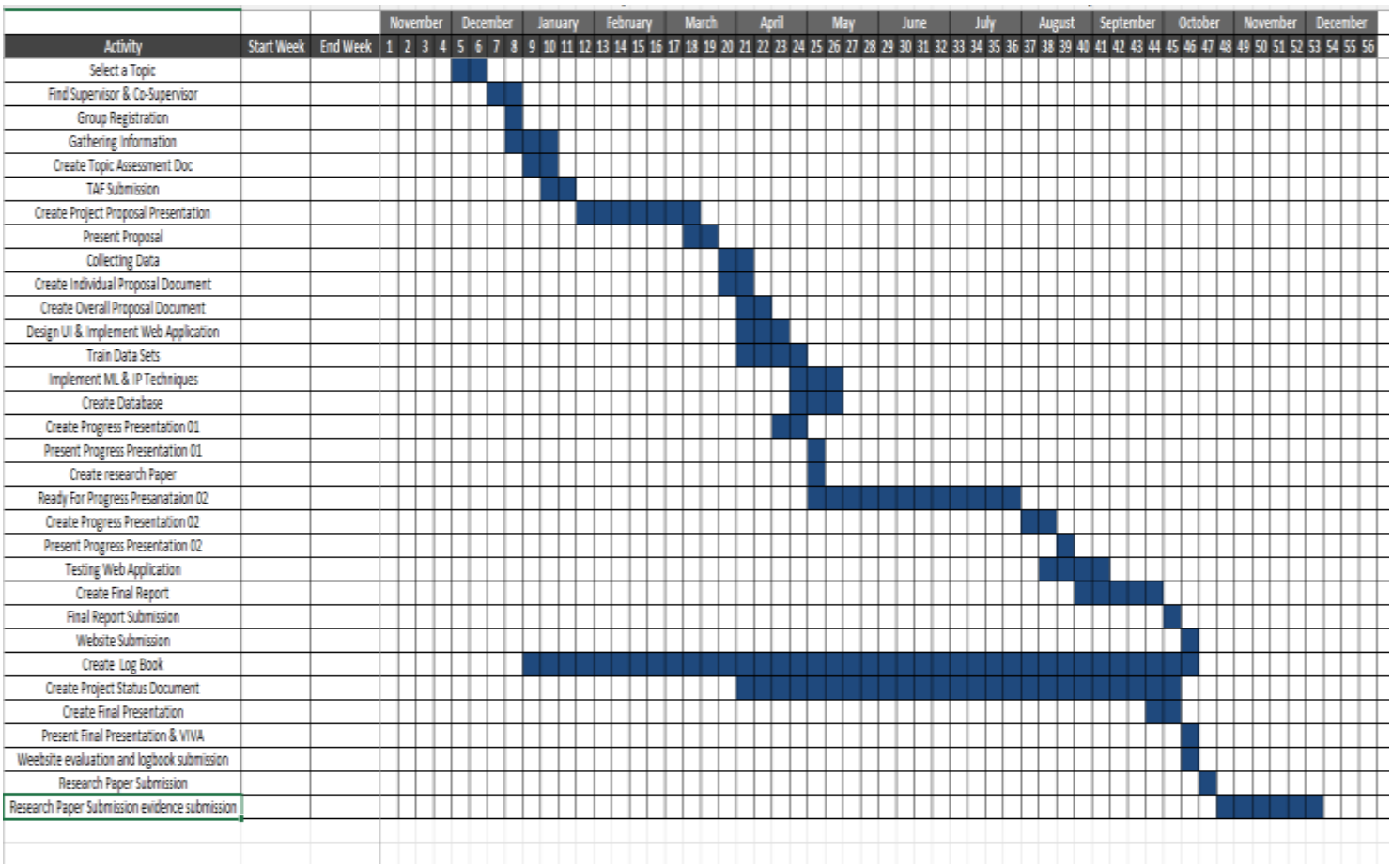



*Figure 6:gantt chart*

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# 10. APPENDICES

Graduate Employability Survey – 2023, Faculty of Management and Finance University of Colombo

University of Colombo GES Report 2022

UGC Tracer Study 2018 Employer Perceptions

Ceylon Chamber of Commerce (CCC) Employer Expectations Survey

ROAR Media Employer Analysis (2019, 2023 data)

https://timesofindia.indiatimes.com/education/news/global-graduates-are-ready-for-work-but-58-of-them-still-cant-find-jobs-whats-holding-them-back/articleshow/122500575.cms

https://fortune.com/2024/10/25/workers-employees-want-soft-skills-eq-upskilling

https://www.newsweek.com/most-recent-graduates-cant-find-full-time-job-their-degree-2128400

*Apendix 1 Survey Results*

The following statements are about how you view yourself. Please indicate to what extent the statements apply to you. Tick the answer of your choice.

Rating scale ranging from I (*not at all like me*) to 5 (*very much like me*).

**13-item Brief Self-Control Scale (Tangney, Baumeister, & Boone, 2004).**

| | | |
|---|---|---|
| 1 | I am good at resisting temptation. | + |
| 2 | I have a hard time breaking bad habits. | – |
| 3 | I am lazy. | – |
| 4 | I say inappropriate things. | – |
| 5 | I do certain things that are bad for me, if they are fun. | – |
| 6 | I refuse things that are bad for me. | + |
| 7 | I wish I had more self-discipline. | – |
| 8 | People would say that 1 have iron self-discipline. | + |
| 9 | Pleasure and fun sometimes keep me from getting work done. | – |
| 10 | I have trouble concentrating. | – |
| 11 | I am able to work effectively toward long-term goals. | + |
| 12 | Sometimes I can't stop myself from doing something, even if I know it is wrong. | – |
| 13 | I often act without thinking through all the alternatives. | – |

*Apendix 2 Self Control Scale*

# PERCEIVED STRESS SCALE

**Sheldon Cohen**

The *Perceived Stress Scale* (PSS) is the most widely used psychological instrument for measuring the perception of stress. It is a measure of the degree to which situations in one's life are appraised as stressful. Items were designed to tap how unpredictable, uncontrollable, and overloaded respondents find their lives. The scale also includes a number of direct queries about current levels of experienced stress. The PSS was designed for use in community samples with at least a junior high school education. The items are easy to understand, and the response alternatives are simple to grasp. Moreover, the questions are of a general nature and hence are relatively free of content specific to any subpopulation group. The questions in the PSS ask about feelings and thoughts during the last month. In each case, respondents are asked how often they felt a certain way.

**Evidence for Validity:** Higher PSS scores were associated with (for example):

- failure to quit smoking
- failure among diabetics to control blood sugar levels
- greater vulnerability to stressful life-event-elicited depressive symptoms
- more colds

**Health status relationship to PSS:** Cohen et al. (1988) show correlations with PSS and: Stress Measures, Self-Reported Health and Health Services Measures, Health Behavior Measures, Smoking Status, Help Seeking Behavior.

**Temporal Nature:** Because levels of appraised stress should be influenced by daily hassles, major events, and changes in coping resources, predictive validity of the PSS is expected to fall off rapidly after four to eight weeks.

**Scoring:** PSS scores are obtained by reversing responses (e.g., 0 = 4, 1 = 3, 2 = 2, 3 = 1 & 4 = 0) to the four positively stated items (items 4, 5, 7, & 8) and then summing across all scale items. A short 4 item scale can be made from questions 2, 4, 5 and 10 of the PSS 10 item scale.

**Norm Groups:** L. Harris Poll gathered information on 2,387 respondents in the U.S.

**Norm Table for the PSS 10 item inventory**

| Category | N | Mean | S.D. |
|---|---|---|---|
| **Gender** | | | |
| Male | 926 | 12.1 | 5.9 |
| Female | 1406 | 13.7 | 6.6 |
| **Age** | | | |
| 18-29 | 645 | 14.2 | 6.2 |
| 30-44 | 750 | 13.0 | 6.2 |
| 45-54 | 285 | 12.6 | 6.1 |
| 55-64 | 282 | 11.9 | 6.9 |
| 65 & older | 296 | 12.0 | 6.3 |
| **Race** | | | |
| white | 1924 | 12.8 | 6.2 |
| Hispanic | 98 | 14.0 | 6.9 |
| black | 176 | 14.7 | 7.2 |
| other minority | 50 | 14.1 | 5.0 |

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# Perceived Stress Scale

The questions in this scale ask you about your feelings and thoughts **during the last month**. In each case, you will be asked to indicate by circling *how often* you felt or thought a certain way.

Name _____ Date _____

Age _____ Gender (*Circle*): **M F** Other _____

| | 0 = Never | 1 = Almost Never | 2 = Sometimes | 3 = Fairly Often | 4 = Very Often |
|---|---|---|---|---|---|
| 1. In the last month, how often have you been upset because of something that happened unexpectedly? | 0 | 1 | 2 | 3 | 4 |
| 2. In the last month, how often have you felt that you were unable to control the important things in your life? | 0 | 1 | 2 | 3 | 4 |
| 3. In the last month, how often have you felt nervous and "stressed"? | 0 | 1 | 2 | 3 | 4 |
| 4. In the last month, how often have you felt confident about your ability to handle your personal problems? | 0 | 1 | 2 | 3 | 4 |
| 5. In the last month, how often have you felt that things were going your way? | 0 | 1 | 2 | 3 | 4 |
| 6. In the last month, how often have you found that you could not cope with all the things that you had to do? | 0 | 1 | 2 | 3 | 4 |
| 7. In the last month, how often have you been able to control irritations in your life? | 0 | 1 | 2 | 3 | 4 |
| 8. In the last month, how often have you felt that you were on top of things? | 0 | 1 | 2 | 3 | 4 |
| 9. In the last month, how often have you been angered because of things that were outside of your control? | 0 | 1 | 2 | 3 | 4 |
| 10. In the last month, how often have you felt difficulties were piling up so high that you could not overcome them? | 0 | 1 | 2 | 3 | 4 |

Please feel free to use the *Perceived Stress Scale* for your research. The PSS Manual is in the process of development, please let us know if you are interested in contributing.


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**References**

The PSS Scale is reprinted with permission of the American Sociological Association, from Cohen, S., Kamarck, T., and Mermelstein, R. (1983). A global measure of perceived stress. *Journal of Health and Social Behavior, 24*, 386-396.


*Apendix 3 Stress Scale*

**The Proactive Personality Scale**

The Proactive Personality Scale is a 17-item measure which measures a personal disposition toward proactive behavior. In this measure, people with proactive personalities tend to scan for opportunities, show initiative, take action, and persevere until they reach closure by bringing about change.

**Categories**

Geographies Tested: United States of America

Populations Included: Female, Male

Age Range: Adults

**Items:**

1. I am constantly on the lookout for new ways to improve my life
2. I feel driven to make a difference in my community, and maybe the world
3. I tend to let others take the initiative to start new projects*
4. Wherever I have been, I have been a powerful force for constructive change
5. I enjoy facing and overcoming obstacles to my ideas
6. Nothing is more exciting than seeing my ideas turn into reality
7. If I see something I don't like, I fix it
8. No matter what the odds, if I believe in something I will make it happen
9. I love being a champion for my ideas, even against others' opposition
10. I excel at identifying opportunities
11. I am always looking for better ways to do things
12. If I believe an idea, no obstacle will prevent me from making it happen
13. I love to challenge the status quo
14. When I have a problem, I tackle it head-on
15. I am great at turning problems into opportunities
16. I can spot a goof opportunity long before others can
17. If I see someone in trouble, I help out in any way I can

**Response Options:**

7- Point Likert scale

Not at all proactive - 1

Extremely proactive - 7

*Apendix 4 : Personality Scale*

Using the scale below, please indicate to what extent each of the following items corresponds to the reasons why you are presently involved in your work

| Does not correspond at all | | | Corresponds moderately | | | Corresponds exactly |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. Because this is the type of work I chose to do to attain a certain lifestyle. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2. For the income it provides me. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 3. I ask myself this question, I don't seem to be able to manage the important tasks related to this work. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 4. Because I derive much pleasure from learning new things. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 5. Because it has become a fundamental part of who I am. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 6. Because I want to succeed at this job, if not I would be very ashamed of myself. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 7. Because I chose this type of work to attain my career goals. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 8. For the satisfaction I experience from taking on interesting challenges | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 9. Because it allows me to earn money. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 10. Because it is part of the way in which I have chosen to live my life. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 11. Because I want to be very good at this work, otherwise I would be very disappointed. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 12. I don't know why, we are provided with unrealistic working conditions. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 13. Because I want to be a "winner" in life. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 14. Because it is the type of work I have chosen to attain certain important objectives. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 15. For the satisfaction I experience when I am successful at doing difficult tasks. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 16. Because this type of work provides me with security. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 17. I don't know, too much is expected of us. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 18. Because this job is a part of my life. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

*Note.* Intrinsic motivation = 4,8,15; integrated regulation = 5,10,18; identified regulation = 1,7,14; introjected regulation = 6,11,13; external regulation = 2,9,16; amotivation = 3,12,17.

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